QMX+ Web Remote Control

Installation and User Guide for ESP32-S3-Zero

Overview

This project is based on the work of Randy Rollins KD7RR, who in February 2026 published the project QMXplus Remote for QMX+ using Aux serial and ESP32, on https://github.com/rrollins72/QMX-Serial-remote

In his original project, a stamp-sized ESP32-S3-Zero microcontroller was combined with a small 1.8" TFT display, a rotary encoder, and a few push buttons to create a compact standalone remote display and controller for the QMX+ transceiver, communicating via the AUX serial port.

In this new project, by Dan Åkerlund SM0EHV and Claude.AI, a WiFi-based web interface has been added, allowing the transceiver to be monitored and controlled from any web browser on the local network — on a desktop or laptop computer, tablet, or smartphone. The web interface mirrors the information shown on the TFT display and provides full remote control of all key functions.

The sketch works both with and without the TFT display, rotary encoder, and push buttons. If you only want the web interface, you simply connect the ESP32-S3-Zero to the QMX+ AUX serial port and power it via USB — no other hardware is required.

What you can see and do from the web interface:

- Real-time display of VFO A and VFO B frequencies, updated multiple times per second
- RX/TX indicator — shows TX in red when transmitting, with SWR value displayed alongside
- S-meter with analog bar graph and numerical dB value
- Local time clock
- Full remote control: change frequency, mode (USB/LSB/CW/DIGI), step size, band, and volume
- Tune buttons (up/down arrows) for all step sizes: 10 Hz, 100 Hz, 1 kHz, 10 kHz
- Mouse scroll wheel tunes the frequency using the currently selected step size
- Band selector with all HF bands from 160m to 6m
- Volume slider

The button in the lower right corner of the page serves two purposes:

- On a desktop computer: opens the page in a compact pop-out window without the browser navigation bar, making it easy to place alongside other programs such as WSJT-X, logging software, or a waterfall display
- On a tablet or smartphone: activates full-screen mode for maximum screen usage

The web interface is accessible via the address http://qmx.local/ on most devices (iPhone, iPad, Android, Windows with Bonjour/iTunes installed). If this does not work, use the IP address shown in the Arduino Serial Monitor at startup.

Hardware Required

You only need three items to get started:

1. ESP32-S3-Zero (Waveshare)

A compact and inexpensive microcontroller with built-in WiFi. Available from AliExpress for approximately $4-5 USD. Search for "Waveshare ESP32-S3-Zero". It is powered directly from the USB-C port.

2. Shielded cable with 3.5mm stereo jack

Used to connect the ESP32 to the QMX+ AUX serial port. A short shielded cable (20-30 cm) is recommended to reduce interference. Connect as follows:

- Tip (top part of plug) → GPIO 17 on ESP32-S3-Zero
- Ring (middle part of plug) → GPIO 18 on ESP32-S3-Zero
- Sleeve (bottom/base of plug) → GND (pin 2) on ESP32-S3-Zero

Available from AliExpress for approximately $1-2 USD. Search for "3.5mm stereo audio cable shielded".

3. USB-A to USB-C cable

Used to connect the ESP32 to your computer for programming, and for power during normal use. Important: use a data cable, not a charge-only cable. Many cheap cables are charge-only and will not work for programming. Available from AliExpress for approximately $1-2 USD.

> **Note:** No display, no rotary encoder, and no extra buttons are required for the web-only version of this project.

Software Installation — Quick Overview

The installation involves the following steps:

1. Install the Arduino IDE
2. Familiarize yourself with the Arduino IDE
3. Download the .ino file and open it in Arduino
4. Install required libraries
5. Install the ESP32 board package
6. Edit the sketch — enter your WiFi credentials and time zone offset
7. Make a backup of the edited sketch
8. Upload the sketch to the ESP32-S3-Zero

Detailed Installation Instructions

1. Install the Arduino IDE

Download and install the Arduino IDE from the official website:

```
https://www.arduino.cc/en/software
```

Choose the version for your operating system (Windows, macOS, or Linux). The installer will guide you through the process. Version 2.x is recommended.

2. Familiarize Yourself with the Arduino IDE

The Arduino IDE has a few key buttons and features you will need:

- Verify/Compile (checkmark icon, top left) — checks the code for errors without uploading
- Upload (right-arrow icon, next to Verify) — compiles and uploads the code to the connected board
- Serial Monitor (magnifying glass icon, top right) — shows text output from the ESP32, useful for seeing the IP address and debugging
- Library Manager (book icon in the left sidebar) — used to search for and install libraries
- Boards Manager (circuit board icon in the left sidebar) — used to install support for different microcontrollers

> **Note:** The first time you compile and upload, it may take several minutes as the IDE downloads and prepares the toolchain.

3. Download the .ino File and Open It in Arduino

Download the file qmxserdisp2_europ_plus_web.ino from the forum. When you double-click the .ino file, Arduino IDE will open it automatically. It may ask you to create a folder with the same name as the file — click Yes.

Note: Arduino requires that the .ino file is located in a folder with exactly the same name. For example, the file qmxserdisp2_europ_plus_web.ino must be in a folder called qmxserdisp2_europ_plus_web.

4. Install Required Libraries

The sketch requires several libraries. Open the Library Manager (book icon in the left sidebar) and search for and install each of the following:

Display:

- ST7735_LTSM by Gavin Lyons

Buttons and encoder:

- ezButton
- OneButton by Matthias Hertel

WiFi and web server:

- ESP_Async_WebServer by ESP32Async
- AsyncTCP by ESP32Async

The following libraries are included in the ESP32 board package and do not need to be installed separately:

- WiFi.h
- ESPmDNS.h

Note: When searching for ESPAsyncWebServer and AsyncTCP, make sure to select the versions by ESP32Async — NOT the older versions by me-no-dev. The me-no-dev versions are not compatible with ESP32 Arduino package version 3.x and will cause compilation errors.

5. Install the ESP32 Board Package

To program the ESP32-S3-Zero, you need to add ESP32 board support to Arduino IDE:

- Go to File → Preferences
- In the field "Additional Board Manager URLs", add the following URL:

```
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
```

- Click OK
- Open the Boards Manager (circuit board icon in the left sidebar)
- Search for "esp32" and install the package by Espressif Systems, version 3.x
- After installation, go to Tools → Board → esp32 → Waveshare ESP32-S3-Zero

You also need to modify one file in the ESP32 board package to correctly map the SPI pins. Navigate to:

```
C:\Users\[username]\AppData\Local\Arduino15\packages\esp32\hardware
\esp32\3.x.x\variants\waveshare_esp32_s3_zero\pins_arduino.h
```

Change the SPI pin definitions to:

```
static const uint8_t SS   = 10;  // FSPICS0

static const uint8_t MOSI = 11;  // FSPID

static const uint8_t MISO = 13;  // FSPIQ

static const uint8_t SCK  = 12;  // FSPICLK
```

Note: This change may need to be reapplied after updating the ESP32 board package, as updates overwrite this file.

Additionally, in the Arduino IDE go to Tools and make sure USB CDC On Boot is set to Enabled. This is required for the Serial Monitor to work correctly.

6. Edit the Sketch

Before uploading, you need to make a few changes to the sketch:

WiFi credentials — find these lines near the top of the sketch:

```
const char* ssid     = "YOUR_WIFI_SSID";

const char* password = "YOUR_WIFI_PASSWORD";
```

Replace YOUR_WIFI_SSID with the name of your WiFi network, and YOUR_WIFI_PASSWORD with your WiFi password. Keep the quotation marks.

Time zone offset — find this line:

```
const int timelocal = 1;     // GMT offset for local time
```

Change the number 1 to your UTC offset. Examples:

- UTC+1 (Central European Time, CET): use 1
- UTC+2 (Central European Summer Time, CEST): use 2
- UTC-5 (Eastern Standard Time, EST): use -5
- UTC-7 (Mountain Standard Time, MST): use -7

Note: The time is received from the QMX+ internal clock, which should be set to UTC. The timelocal offset converts it to your local time for display.

Band start frequencies — optionally, you can customize the frequency that each band button tunes to. Find the bandsets array:

```
const char bandsets[11][12] = {
  "00001836000",  // 160m
  "00003560000",  // 80m
  ... and so on
};
```

Each frequency is 11 digits, in Hz. For example, 14.074 MHz (FT8 on 20m) would be written as 00014074000.

7. Make a Backup

After editing the sketch with your WiFi credentials and time zone, save a backup copy of the file. This is especially important because the WiFi password and your personal settings would otherwise need to be re-entered after any future updates to the sketch.

Simply copy the entire sketch folder to a safe location, for example:

```
qmxserdisp2_europ_plus_web_MY_SETTINGS
```

8. Upload the Sketch to the ESP32-S3-Zero

Follow these steps to upload the sketch:

- Hold down the BOOT button on the ESP32-S3-Zero
- While holding BOOT, connect the USB-C cable to your computer
- Release the BOOT button
- In Arduino IDE, make sure the correct board is selected: Tools → Board → Waveshare ESP32-S3-Zero
- Select the correct COM port: Tools → Port → (select the port that appeared when you connected the ESP32)
- Click the Upload button (right-arrow icon)
- Wait for the upload to complete — this may take 1-2 minutes the first time
- After uploading, disconnect and reconnect the USB cable to restart the ESP32 normally

After restarting, open the Serial Monitor (magnifying glass icon) and set the baud rate to 115200. You should see output similar to:

```
Connecting to WiFi...

WiFi connected! IP: 192.168.0.78
```

```
mDNS started: http://qmx.local/

Web server started
```

Note the IP address shown — you can use this to access the web interface directly if http://qmx.local/ does not work on your device.

Connecting to the QMX+

The ESP32 communicates with the QMX+ via the AUX serial port. You need to enable and configure this port on the QMX+ using a USB connection to a computer:

- Connect the QMX+ to your computer via USB
- Open a terminal program (or use the Arduino Serial Monitor) at the correct baud rate
- Navigate the QMX+ menu: Main menu → Configuration → System config → GPS & Ser. Ports
- Set Serial 1 on AUX: ENABLED
- Set Serial 1 baud: 19200

Note: Higher baud rates were tested but resulted in missed data. 19200 baud has proven reliable.

Accessing the Web Interface

Once the ESP32 is powered and connected to your WiFi network, open a web browser and go to:

```
http://qmx.local/
```

This works on most modern devices. If it does not work, use the IP address shown in the Serial Monitor at startup, for example:

```
http://192.168.0.78/
```

mDNS compatibility by device type:

- iPhone and iPad: works automatically
- Android (modern): works on most devices
- Android (older): may not work — use IP address instead
- Windows: works if Bonjour is installed (comes with iTunes)
- macOS: works automatically

Note: For the most reliable access from all devices, bookmark the IP address. To always get the same IP address, configure a DHCP reservation for the ESP32 in your router settings using the MAC address shown in the Serial Monitor.

Using the Pop-Out Window on Desktop

When using the web interface on a desktop computer alongside programs like WSJT-X, clicking the expand button (lower right corner) opens a compact pop-out window without the browser navigation bar. This window can be resized and placed anywhere on your screen or on a second monitor.

The mouse scroll wheel can be used to tune the frequency up and down using the currently selected step size. This makes it easy to tune quickly without clicking the arrow buttons.

Troubleshooting

Compilation error: Serial2 not declared

Make sure the correct board is selected: Tools → Board → Waveshare ESP32-S3-Zero. Serial2 is only available on ESP32 boards.

Compilation error related to mbedtls or tcp_state

You have the wrong version of ESPAsyncWebServer or AsyncTCP installed. Remove the me-no-dev versions and install the ESP32Async versions instead.

Cannot find COM port

Make sure you held the BOOT button when connecting the USB cable. Also try a different USB cable — charge-only cables will not work.

"No contact with QMX+" shown on web page

Check that the 3.5mm cable is correctly connected (tip to GPIO17, ring to GPIO18, sleeve to GND). Also verify that the QMX+ AUX serial port is enabled at 19200 baud.

http://qmx.local/ does not work

Use the IP address instead (shown in Serial Monitor). On Windows, install Bonjour (comes with iTunes).

Web page shows correct frequency but band button is wrong

This resolves itself after the first frequency update from the QMX+. The band is automatically detected from the active frequency.